

Global Facilities - Design & Construction Standard - Construction Execution Plan

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1.0 Purpose

- 1.1 The Construction Execution Plan (CEP) is prepared to describe how Micron Project Construction Management Team (MPMT) will follow up and monitor the construction, completion, commissioning, and handover of the contract objects awarded to xx – contractor xx.
- 1.2 The CEP documents how Micron Project Team ensures the project is executed in compliance with Micron requirements.
- 1.3 This includes but is not limited to describing the organization of the Micron xx project Team, objectives, execution strategies/processes, success factors and interaction with xx contractor xx as well as other groups in the Micron Site Team in xx location xx.

2.0 Background

- 2.1 The contract for the xx FAB, xx – (Contractor name). As (EPC / Design build contractor xx -Contractor name- xx) has full responsibility for the xx *Example* -Engineering, Procurement and Construction- xx work up to and including xx scope up to handover xx. Micron is responsible for xx State Micron responsibility xx – (Example may be: Tool Install / H/U with specialty third party contractors.)

- 2.2 The work will be undertaken in the xx location xx facilities in xx Country xx- area- xx.
- 2.3 The overall project execution strategies are described in the "Project execution plan", document xx-xxx – doc number. The xx project) entails engineering, procurement, construction, commissioning at multiple locations in addition to xx – site location xx, for project components such as the xx – examples – Chillers / Air handling units etc. etc. -xx. The construction plans, and strategies described in this document are part of a larger whole, and applicable learnings will be shared across the project in the construction area.
- 2.4 Micron has experience working with xx – contractor name xx on previous projects including xx project name xx Micron experiences from these projects have been used to help in the development of this Project specific Construction Execution Plan.

3.0 Scope

- 3.1 The scope of this CEP is to all Micron sites and targeted to all Micron team members working on construction projects within Micron sites.

4.0 Definitions and Acronyms

- 4.1 CEP: Construction Execution Plan
- 4.2 CR protocol: Cleanroom protocol
- 4.3 EHS: Environment, Health & Safety
- 4.4 FCT: Facilities Central Team. This is used interchangeably with Global Facilities Engineering Team and associated to O&M BKMs
- 4.5 EPC: Engineering, Procurement and Construction
- 4.6 GFTT: Global Facilities Technology Team. This is used interchangeably with Global Facilities Engineering Team and associated to design standards ITP: Inspection & Test Plan
- 4.7 GFC: Global Facilities & Construction Team. This is used interchangeably with Global Facilities (GFac), Facilities Central Group (FCG), World Wide Operations (WWOps), World Wide Facilities (WW_FACILITIES), Corp Facilities
- 4.8 Global Facilities Central Team: This is used interchangeably with Innovation & Integration Team
- 4.9 Global Facilities Construction Team: This is used interchangeably with Global Construction Team
- 4.10 Global Facilities Operations Support Team: This is used interchangeably with Global Facilities Operations Team
- 4.11 Global Facilities Planning Team: This is used interchangeably with Global Facilities Line Planning Team

- 4.12 IIF: Injury Incident Free
- 4.13 KPI: Key Performance Index
- 4.14 MPMT: Micron Project Construction Management Team
- 4.15 MPP: Manpower plan
- 4.16 NCR: Non-Conformance Report
- 4.17 PMT: Micron Project Management Team
- 4.18 QA: Quality Assurance
- 4.19 QAI: Quality Action Issue report
- 4.20 QC: Quality Control
- 4.21 QM: Quality Management
- 4.22 RFE: Ready for Equipment
- 4.23 RFM: Ready for Manufacturing
- 4.24 WSR: Weekly Surveillance Report

5.0 Objectives for Micron Project Site Team

- 5.1 The objectives for the Micron Project site team for xx – Project name - xx are as follows:
- 5.1.1 Provide active and documented contribution and support to the Project and Corporate EHS Objectives
 - 5.1.2 Plan and execute the construction follow-up activities on site in compliance with the Micron guidelines and contract requirements.
 - 5.1.3 Ensure project objectives related to EHS, progress, quality and cost are met through the following:
 - Proactive and documented involvement in site EHS (EHS Plan, safety inspections, IIF programs etc.)
 - Early, systematic and efficient involvement in design development (constructability & design reviews)
 - Define and use efficient KPIs for monitoring and reporting of status trends on progress and quality of works
 - Efficient implementation of corrective actions as and when required
 - Be recognized as a pro-active and cooperative group within the Micron Site Team
- 5.2 The overall objectives described above shall be achieved via cooperation with the xx contractor xx and by setting up an organization that is tailored to the strategy for successful completion of the contract objects at site.
- 5.3 Some key activities in the execution plan include the following:
- 5.3.1 Development of construction organization, staffing plan and associated job descriptions defining key responsibilities
 - 5.3.2 Development of Site Mobilization Plan Identify and focus on Key Construction Activities and construction Input Requirements for the following;
 - EHS Requirements for Construction Execution
 - Constructability Plan and Execution
 - Construction Cost Estimate and Schedule Planning
 - Construction Method and Sequence
 - 5.3.3 Key Construction Deliverables (Engineering & Procurement)
 - Implementation of lessons learnt from past projects
 - Identification of risks and corresponding mitigation measures (barriers and barrier breaking strategies)
 - Identification of necessary constructability workshops

and reviews for each phase

- Development of checklists and work instructions as required to perform key activities systematically and, in a manner, consistent with Project requirements and good construction practices.
- Establishment of attendance plan for design reviews, construction methodology/assembly reviews, planning reviews, **Risk assessment**, Tool move in, Hook up, etc.
- Establishment of attendance requirements for inspections and tests by mark up of Hold & Witness points in Quality Management ITP.

5.3.4 Objectives are broken down in main activities and are listed as part of the Construction Execution Activity Plan in **Section 10** of this document.

6.0 Project Summary

6.1 General

6.1.1 Once completed the xx project will be located xx describe location and over view of the project example size, capacity, output etc. etc. (Possible examples below)

- XX FAB and CUB
- Admin building
- Canteen for xx amount of employees

6.2 Project description

6.2.1 Description of the xx FAB/CUB should include the following:

- E.g. The xx and supporting Admin building will be concrete and steel structure with
- Capacity of the project
- No of chillers
- Size of WWT plant , Gas farm / Chems etc.
- Key aspects of the project that will form a key to its success, e.g. The attention in ensuring the project follows a CLEAN DRY BUILD approach – 5 stages of construction etc. plus other factors
- Mention of the major objects that the xx FAB comprises: (typical example)
 - FAB
 - CUB
 - Admin building

- Canteen with xx capacity
- Car park with space for xx cars / motorcycles

6.3 xx Contractor name xx

6.3.1 This CEP is to a great extent based on Microns past experience with xx Contractors name – location xx.

6.3.2 Considerations are also made around the fact that xx Contractor name xx has basically limited experience as an EPC (Engineering Procurement Construction Contractor); engineering and procurement shall now be performed by xx contractor name xx and not be a Client delivery. In order to understand the challenges associated with executing an EPC contract at xx contractors name xx, and to set up the Micron project Team correctly, it is vital to recognize the main strengths and weaknesses associated with construction at xx Contractors name / location xx.

6.3.3 Xx Contractors name xx Strengths

- Xx Contractors name xx have the following strengths when it comes to performing the Construction and Completions work on this project: (Typical examples given below)
 - They appear to have unlimited construction resources at the work force level. If they need more fitters, welders or painters, they can readily source them from their pool of available sub-contractors.
 - They are prepared to listen and take advice from the Micron team, especially if a good relationship is established early on.

6.3.4 Xx Contractors name xx Weaknesses

- Xx Contractor name xx have the following weaknesses when it comes to performing the Construction, Commissioning and Completions work on this project:
 - Lack of full Micron engineering, procurement and construction experience and understanding
 - Limited construction supervision personnel and those they have are not always the best due to the limited compensations that they normally offer. Experience has shown that the best of the xx contractors name xx Project management team will move on to other companies for better pay.
 - Not aware of the full implication and requirements related to Clean protocol and the Clean dry build

method during the various stages of the project, this may hamper their usual methods of working and may create frustration. Lack of Microns Clean and dry building philosophy requirements to be understood and fully implemented by all relevant sub-contractors

- Execution based on sub-contracting philosophy, very limited use of internal resources.
- Lack of ability to establish and follow an agreed and approved plan. The subcontractors will be trying to complete their work as fast as possible in the order that generates the most revenue to them. They will tend to leave the awkward and complicated bits to the end and may be working out of sequence. It is generally the external pressure from the Micron team that gets them working in the right sequence. In some cases (depending on geographical location) the subcontractors tend to have more 'respect' for the client (Micron) team than the xx Contractors name xx project management team which is leading to an increased demand for client construction resources.
- Xx Contractors name xx Quality team is limited in size and is always stretched leading to mistakes or missing out on crucial checks. It is the Client team that will normally generate the most NCR's on the project.
- Limited understanding of the commissioning and Completion system, especially if it needs to adhere to any Specific Micron requirements.

7.0 Organization – Micron Construction Team

7.1 Strategy

7.1.1 The xx Project name xx Project Manager is responsible for setting up the construction team organization and establishing and maintaining a manpower mobilization plan for the execution phase.

7.1.2 The basis and strategy for construction manpower planning are listed below:

- In order to ensure an efficient utilization of Micron resources, and to secure successful delivery of the contract objects, a fully aligned Micron project Team, including engineering, procurement, construction and Quality management, is required.
- During the detail design and fabrication execution phases, Construction lead positions, Project Manager, PMO, GFTT, EHS, procurement, will be mobilized to follow up planning, constructability and ready for construction activities.
- During the fabrication phase, additional construction support positions like Discipline leads, Supervisors, Materials & Logistics Controller, Planner, Contracts
- Administrator and Quantity Surveyors will be filled to support the execution of fabrication work.
- In addition to the Micron Site project team, discipline Quality inspectors may be mobilized at various stages or throughout the duration of the project by Micron QM team to carry out surveillance, oversight, and inspections at fabrication sites, including surveillance activities at sub-contractors and vendors facilities. Note that such services may be conducted through calling off services from local inspection service companies.
- During construction, and when detail engineering activities diminish and follow-up engineering support for fabrication picks up, Micron discipline engineers will be used as support to construction and to participate in Completion and commissioning activities. The intent is to always utilize Micron engineers where it is most beneficial for the project.
- A position in the Micron Project team may be given the responsibility to handle hook-up preparations of all contract objects.
- A good relationship with the opposite numbers in xx Contractors name xx Construction organization is identified as an important success factor in the project. Timeline for mobilization must reflect this and allow

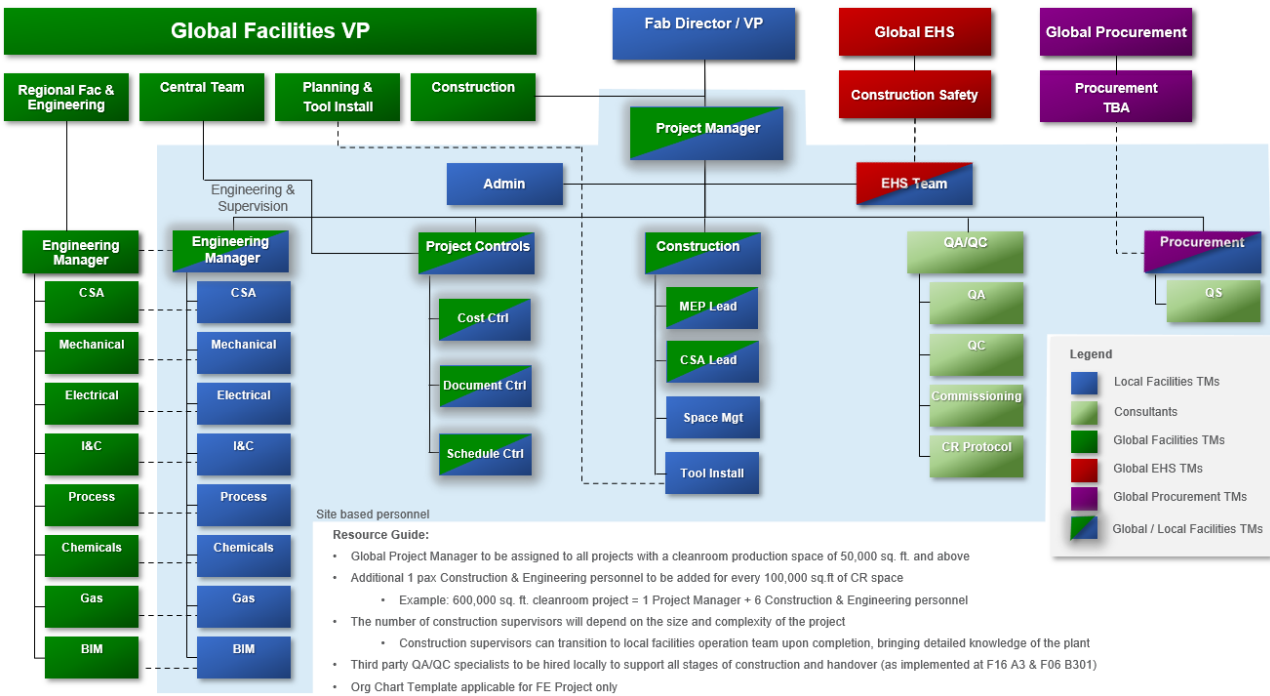
adequate time to establish this relationship during preparations and planning for construction.

7.2 Organization

7.2.1 The xx Project name xx Project Team will be headed and managed by a Project Manager and the top level of this organization as shown in **Figure 1**.

7.2.2 The Micron Project Management team (PMT) will generally be manned up with personnel with recent and relevant experience with working in xx specific location / country xx and with xx Contractor name xx. Knowledge about sub-contractors and the way xx contractors name xx is executing construction works is considered vital in order create an efficient follow-up process at the xx Project name xx site.

Figure 1: Standard Project Organization Structure



- 7.2.3 The Construction Team is managed by discipline leads for the various disciplines with Construction Supervisors being responsible for various disciplines.
- 7.2.4 The Construction Supervisors / **Project Engineers** will be supported technically by the relevant discipline design engineers who report to the Engineering Manager.
- 7.2.5 The GFTT Leads shall, in addition to supporting Micron involvement in the Commissioning process, perform the following:
- Follow up on engineering issues that have a direct bearing on construction.
 - Be involved in and coordinate the constructability review process and other issues within construction that require cross functional resolution.
- 7.2.6 For each construction discipline, Quality inspector(s) may be mobilized as and when required, to work closely with the respective Construction Supervisors.
- The Quality Inspectors will report directly to Micron Quality Lead, with a close working relationship with the Construction discipline personnel.
 - The above organization setup mirrors the xx Contractors name xx Project Organization where Construction and Quality reports to the xx Contractors name xx Project Manager / Director.
- 7.2.7 The Micron Project Management Team (PMT) may establish a QM Team, headed by a **QC** Construction / Engineering Lead to pay special attention to the following an from early stage of project execution.
- Ensure strong QM – QA / QC / Commissioning / CR protocol execution throughout the project execution
 - Monitor performance during the entire process
 - Ensure efficient handover to Micron Facilities group
 - Assist xx contractors xx in preparing for and executing completion and handover.
 - Construction Supervisors, Quality Inspectors and Commissioning Engineers will all be actively involved in the QM process, from preparations and planning through field inspection, punch and acceptance of systems.
 - The **QC**/Construction Engineering Lead shall mobilize early in order to perform oversight of the preparation of QA/QC -related procedures. This information is vital in order to build a level 5 construction schedule that is fully aligned with the Facilities Equipment move in schedule. (RFE / RFM milestones)

- 7.2.8 The PMT will also take a proactive role with respect to EHS on site and will liaise closely with both Micron and xx Contractors name xx EHS Depts.
- 7.3 Contractors' responsibilities **(Propose to go under 8.0 Roles and Responsibilities)**
- 7.3.1 The construction execution strategy is to a great extent dictated by the chosen contractor's construction methods and their subcontracting strategy.
- 7.3.2 The contractor, i.e. xx contractors name xx, has full responsibility for the Engineering, Procurement and Construction work including commissioning.
- 7.3.3 The responsibility is transferred to Micron after Final Completion and Handover, but the contractor will continue to provide support services and personnel resources on an as needed basis.
- 7.3.4 Faults identified during commissioning that are attributable to fabrication errors will be returned to contractor for correction at xx contractors name xx cost.
- 7.3.5 The xx contractors name xx shall issue a Construction Method Statement Document describing all major objects, assembly and installation sequences. This construction/assembly method statement document shall also include fabrication sub-contracting, lifting/ strategies, and equipment move in strategies.
- 7.3.6 The xx contractor name xx is also responsible for the following:
- Management of the entire construction process.
 - Develop a detail Level 5 Plan to reflect the Construction Method Statement Document.
 - Ensure correct Engineering and Procurement dependencies are linked into the detailed Level 5 Plan.
 - Issue a Constructability Plan to ensure Fabrication involvement in the design in an ongoing process as per requirements in the contract document xx, e.g. Exhibit xx - contractors name xx, scope of work xx
 - Establish Detail Procurement Schedule, including completion / commissioning Input. (Format to be used shall be in accordance with xx Example - "Procurement Milestone Schedule" attached in Exhibit xx – xx
 - Identify long lead Material and Equipment and the associated Site Need Dates as dictated by the xx Contractors name xx Construction Plan.

- Establish and implement a procedure and a system for Clean and dry building in accordance with Microns Clean room protocol procedure – xx Document number xx during the construction phase.
- Implement Construction and Completion requirements, e.g. requirements for Planning, Procurement/Material Control, Clean Room Protocol, Quality, commissioning, etc.
- Secure construction drivers by active involvement during design and by assisting the design team as required.
- Planning and progress reporting for the EPC process
- Production of the required construction documents, drawings and procedures
- Receiving, storing and preservation of materials and goods
- Prefabrication and construction erection/assembly of contract objects, including quality activities such as visual inspections, NDE (Non-destructive examination) of welds, etc.
- Dimensional control of all contract objects, including co-operation with the Tool install vendors.
- Weekly General Inspection focusing on EHS, Workmanship and Housekeeping.
- Weekly Inspection focusing on Surface Protection
- Weekly Inspection focusing on Clean Room protocol
- Perform preservation checks of all equipment and systems.
- Document quality of end product.
- For further and complete description of xx Contractors xx responsibilities and micron requirements see Contract documents, xx Example – Exhibit A Scope of work. Xx

8.0 Roles and Responsibilities

8.1 Micron Project Manager

- 8.1.1 The xx Project Manager is responsible to the **Project Director** / Director – Global Construction for ensuring that the CEP document is updated promptly based on organizational needs or contractual conditions.
- 8.1.2 The xx Project Manager is responsible for setting up the construction team organization and establishment and maintaining manpower mobilization plan for the execution phase.
- 8.1.3 The xx Project Manager is responsible for planning, execution and completion of the construction / commissioning scope of work. Key responsibilities during organization, planning, execution include the following:
 - Develop, issue, and ensure implementation of Construction Execution Plan (CEP) specifically tailored for xx Project
 - Ensure constructability reviews are completed for all key activities Review work packages, installation methods and schedule
 - Liaise with the Director – Global construction in developing, maintaining, a updating the MPP including quarterly reviews to ensure that the correct manning levels are maintained throughout the project life.
 - Liaise with xx Project EHS Manager in developing EHS Plan
 - Liaise with xx Project Quality Group in developing Quality Plan and Audit and Verification Plan for xx FAB
 - Participate in **Risk** workshops as required.
 - Review the need for Temporary Equipment, establish and maintain a Temporary Equipment Register. and ensure location/interface with xx Facility team.
 - Participate in schedule reviews and provide input to project risk register.
 - Liaise closely with general contractor to ensure project progression in accordance with approved plan.
 - Manage day to day stakeholder relationships and issues.
 - Approve minor variations to budget, schedule and scope, within agreed **parameters**.
 - Maintain a project Risk register and the integration of risk treatments and control activities into the project plans and activities.

- 8.2 Construction Discipline leads for each main discipline group.
- 8.2.1 Construction Discipline leads for each of the main discipline groups will report to the Project Manager and be responsible for the following:
- Ensuring that all specific discipline issues that prevent planned progress are resolved.
 - Ensuring that the necessary quality standards achieved
 - Seek necessary clarifications from the Micron GFTT Engineers relating to specific Micron requirements.
 - Deal on a day-to-day basis with the xx Contractors name xx Project Site / Discipline Supervisors.
 - Attend weekly meetings with the Micron project manager to seek assistance at a higher level with those issues that cannot be resolved in a speedy fashion.
 - Work closely with the QM lead and the respective Micron Quality Inspectors and proactively try to resolve arising quality issues as early as possible with minimum impact on the project schedule.
 - Discipline leads will be responsible to monitor the Constructability process and coordinate Micron discipline individual's involvement in this process as well as design reviews.
- 8.2.2 Discipline personnel will be mobilized as required to support the Discipline leads.
- The CSA, Mechanical and Electrical disciplines have been identified as disciplines where additional resources monitoring installation works are typically required.
- 8.3 Quality
- 8.3.1 The QM Lead shall report to the Project Manager and be responsible for the following:
- Ensure construction activities are planned and coordinated,
 - All QC documentation is produced (prepared and completed) by xx contractor name xx in a format that is suitable for Micron Global Quality system.
 - Attend all the xx Contractor name xx QM meetings and assist them to determine system priorities.
 - Ensure that xx contractor name xx follow up all punch list items to ensure that they are completed in a timely manner prior handover to the facilities group.
- 8.3.2 The opposite number for the QM Lead will be the xx-contractor name xx QM Manager.

- 8.4 Opposite number in the xx Contractors name xx team.
 - 8.4.1 Typically, every role in the Micron PMT should have at least one opposite number in the xx Contractors name xx team to work closely and effectively throughout the project duration.

9.0 Construction Team Mobilization Plan

- 9.1 The need for Micron Construction personnel and their mobilization dates are continuously being monitored and assessed against xx Contractors name xx current schedules and performance.
- 9.2 The current Construction Team Mobilization Plan can be found in the xx Project name xx Manpower plan (MPP), see links below: Link to Project specific mobilization plan.
- 9.3 Project Familiarization
 - 9.3.1 A project familiarization package be prepared by the Project Manager / PMO and given to all new construction team members upon mobilization.
 - 9.3.2 The package shall describe the project, project goals and objectives and refer to all relevant project specifications such as functional requirements, contract strategy, contract management plan, site work instructions and other pertinent documents.
 - 9.3.3 The familiarization shall also include the following information:
 - Site layout and location of project specific fabrication shops /areas
 - Updated project schedule.
 - Update on project status
 - Contract awareness
 - Contract governing documents procedures and standards.
 - 9.3.4 The Project familiarization/induction presentations can be found on the share point folder. Once established – xx link to Project familiarization document xx

10.0 Construction Site Management Plan Team Mobilization Plan

- 10.1 A Construction Site Management Plan must be tailor-made for the project as well as the location where construction activities are carried out.
- 10.2 Critical Site Requirements assessed during bid evaluation and covered by xx contractors name xx under the contract.
- 10.3 Construction Site Management Plans and Procedures This is generally covered by xx contractors name xx under the Contract. Plans and procedures specific to Micron Site Team described below.
 - 10.3.1 Construction (see also section 8 of this document for more details)
 - Constructability and Construction Work Plans developed and maintained by xx contractors name xx , Construction team involved in reviews and acceptance.
 - Roles and responsibilities with Micron PMT defined.
 - Meeting schedule established and maintained.
 - Site coverage plan established and maintained.
 - 10.3.2 EHS
 - Common EHS Program established, safety objectives and strategies defined and communicated.
 - EHS induction and training program agreed and rolled out, focus on cultural differences, objectives and tool (IIF)
 - Process for approving personnel (mainly electrical, instrument and pipe fitters) being developed.
 - Plan for Safety meetings and PMT (xx Contractors name xx and Micron) EHS personnel established.
 - 10.3.3 Engineering
 - Xx Contractors name xx responsibility under the contract, a Micron GFTT Team mobilized to monitor and verify engineering performance.
 - 10.3.4 Project Controls
 - xx Contractors name xx responsibility under the contract, a Micron Project Controls Team mobilized to monitor and verify performance.
 - Requirements regarding DCC and reporting requirements (cost and schedule) are covered under the contract.

10.3.5 Quality Assurance and Control

- Micron Audit Plan developed and implemented.
- Lessons learned process initiated and ongoing. Construction Team facilitating the process.
- Xx Contractor name xx is responsible for Quality Control,
- The Quality Plan (ITP = Inspection & Test Plan) is established for relevant disciplines. Micron (Construction and Quality with input from Engineering) will review and accept.
- Quality related site procedures are established as required.
- Commissioning being handled by xx contractor name xx in close cooperation with the Micron project team.

10.3.6 Procurement

- Xx Contractor name xx is responsible for procurement under the Contract.
- Micron has mobilized resources within both engineering and procurement to support xx Contractor xx name in the entire procurement process, including establish and follow up of the SQS plan

10.3.7 Site Administration

- Local Administration Assistants, with past experience with both Micron and xx Contractor name xx, are mobilized to ensure all aspects of non-technical issues are managed and organized in an efficient manner.

- 10.4 The Micron PMT will continuously monitor the need for additional procedures and/or equipment/tools in the project. Where shortfalls are identified, responsibility for rectifying the shortfall will be assigned by the Project Manager.

11.0 Construction Execution

11.1 General

- 11.1.1 The purpose of this section of the CEP (Construction Execution plan) is to describe the Contractors responsibilities related to construction and completion (high level) and to identify key Micron activities related to construction execution and follow up.
- 11.1.2 The section shall also briefly describe methodology and responsibility for accomplishing these activities.
- 11.1.3 Where required, specific Site Work Instructions will be prepared to describe in more detail how specific tasks and processes shall be executed by the Micron project team.

11.2 EHS Performance

- 11.2.1 EHS shall receive the highest attention from both the Micron Project Team and the xx contractors name xx teams throughout the entire project execution phase at xx location xx Requirements for EHS management during fabrication, construction and commissioning are well defined in the contract with xx Contractor name xx.
- 11.2.2 It is one of the key responsibilities of the Micron Project team representatives to ensure that all EHS requirements are implemented, as well as mentoring and encouraging the whole project team in establishing and maintaining a good EHS culture.
- 11.2.3 In addition, the Micron project team includes an EHS Lead who will support and assist the Project representatives with EHS matters.
- 11.2.4 The Micron EHS Lead is also responsible, together with the contractor(s), for setting up an EHS training plan/matrix and arrange mobilization courses/workshops for all contractors and company personnel working at the site(s). The EHS training matrix/plan must cover all projects and project phases.
- 11.2.5 Visibility on site and participation in EHS related meetings are leading indicators for Micron PMT and PMT EHS performance. Participation of Micron personnel in toolbox meetings, permit meetings and EHS meetings are planned and documented in a Site EHS Coordination Plan which will be updated and displayed on a weekly basis.

11.3 Lessons Learned

- 11.3.1 Lessons Learnt from previous projects by both Micron personnel and/or xx contractors name xx are an integral part of the Constructability Plan for this Project.
- 11.3.2 An extensive review of previous projects of a similar nature shall be carried out and Workshops shall be carried out both within the Micron team and also together with the xx contractors name – PMT xx team to come up with a comprehensive list of Lessons Learnt.
- 11.3.3 The critical success factor for Lessons Learned activities is the openness and willingness of all parties to accept a certain level of criticism which in some cases may be of a personal nature.
- 11.3.4 A list of lessons learnt shall be implemented at the commencement of the execution phase.

11.4 Identified Success factors xx Typical examples xx

- 11.4.1 The following success factors have been identified from Lessons Learnt Workshops:
 - Build on contractor Safety Culture and establish/develop a strong Micron / contractor EHS Partnership
 - Have discipline workshops at the start of the project to highlight specification requirements.
 - Assist Contractor in building a realistic and detailed project schedule.
 - Agree on a robust procedure for selection and Micron acceptance of Contractors subcontractors.
 - Early definition of Completion / commissioning process, including acceptance documents and priorities
 - Purchase of the best equipment/materials, good follow-up of vendor/supplier's performance
 - Avoid high turnover of contractor as well as Micron personnel.

11.5 Constructability Plan and Strategy

11.5.1 Purpose of Constructability Reviews

- The purpose of performing Constructability Reviews is to ensure that all known and relevant construction techniques and methodologies, to ensure a safer, cost effective and operable facility are delivered on time and on budget, are implemented from Concept through to handover of the completed project to Facilities.

- To facilitate this, it is imperative that good communication and interaction is consistently maintained between Engineering and Construction. Whenever a new group enters the different phases of a Project, there are new opportunities to improve on the constructability of the project. Wherever possible the Micron PMT will access the Micron Network of Excellence (NoE) to investigate if anyone within Micron has any new ideas relating to this type of construction activity.

11.5.2 Strategy during the Execution Phase of the Project

- Xx contractor name xx shall assist Micron in establishing and implementing a Constructability Plan specific to xx Project name xx. Recognizing xx contractor name xx may have minimal experience in early involvement in design, the Micron PMT needs to work closely with xx name of contractor xx in developing and implementing the Constructability Plan.
- In order to secure xx contractors name xx commitment to the constructability process the Constructability Plan is prepared and maintained by xx contractors name xx with input/assistance from Micron project Team.
- After conducting a Lessons Learnt Workshop, a Suggestions Log shall be developed to identify those areas where construction efficiency can be improved. The way to improve construction efficiency may be to change the design, engineer things earlier than originally planned, special procurement requirements and also to change the sequence of construction activities. The Suggestions Log also looks at any earlier suggestions made from the earlier conceptual parts of the project and see if they can be incorporated.

11.5.3 Implementation

- Regular meetings need to be held between the Construction group and the other groups that have been identified in the Suggestions Log, such as Engineering and Procurement. The purpose of these meetings is to get the 'buy-in' from these groups to do things differently to assist construction.
- In addition, there may be occasions to have specific meetings between all groups to identify the pros and cons if a "new" build method is identified. Implementation of the constructability plan is a continuous improvement process.

11.6 Contractor Construction Planning, incl. Fabrication Method Statement

- 11.6.1 xx Contractor name xx is responsible to develop a detailed Fabrication Method Statement describing the sequence of fabrication and MEP outfitting in detail. This document will form the basis for the Master schedule including engineering. Procurement activities shall be scheduled to support the sequence described in the fabrication method statement.
- 11.6.2 The Micron Project team shall work closely with xx contractors name xx construction PMT to establish a safe and efficient fabrication sequence and will also initiate and perform workshops to assess and optimize the MEP outfitting schedule for the various areas in the FAB and CUB buildings.
- 11.6.3 The Project Manager is responsible for coordinating and performing these activities within the Micron Project Team.
- 11.6.4 Major Construction Milestones
 - The main construction milestones in xx contractors name xx Project execution program should be listed.

11.7 Construction Project Controls

- 11.7.1 The Project Discipline leads along with the PMO are responsible to perform and document these activities within the Micron Project Team.
- 11.7.2 Accurate reporting of progress and status is of paramount importance to ensure efficient management of a project. Although xx contractors name xx is responsible for progress and status reporting, verification of status by Micron is required. The Micron Project Team shall systematically assess status of the Engineering, Procurement and Construction activities performed by xx name of contractor xx and report any findings to the Micron Project Control Team.
- 11.7.3 Typically, a dedicated Construction Planner / PMO shall be assigned to the Micron Project team and he will, independently of xx contractors name xx, carry out progress measuring audits to ensure that xx contractors name xx are neither over nor under estimating the actual achieved progress. He will also assist the Construction Discipline Groups to investigate required changes to the construction sequence to determine what may be the overall impact to the complete construction schedule. Alternatives will be evaluated to ensure that an optimum way forward is always chosen.

- 11.7.4 Where slippages/discrepancies to the approved Master schedule are identified possible mitigating actions shall be identified and discussed internally and thereafter with xx contractors name xx.
- 11.7.5 Micron requirements regarding project controls are described in xx refer to Exhibit xx of the contract.
- 11.8 Key Performance Indices (KPIs)
 - 11.8.1 The Project Construction Team shall identify and implement a set of KPIs, other than the traditional KPI related to quality, and to assist in ensuring that there is no gap between planned and actual progress. The subject KPI shall be discussed and agreed with xx name of contractor xx and shall be followed up on a regular (weekly, bi-weekly or monthly) basis.
 - 11.8.2 Examples of KPIs include the following:
 - Revisions to drawings and documents after IFC issue.
 - Number of HOLDS on drawings issued each week combined with number of HOLDS released each week
 - Tag generation and population into *Tag Register (* Suggestion)
 - 11.8.3 The PMO will be responsible for following up and reporting the KPIs within the Micron Construction team.
- 11.9 Construction Site Mobilization Plan
 - 11.9.1 Xx Contractor Name xx is responsible for establishing a Site Mobilization Plan which describes activities (preparation of facilities, mobilization of Construction and Quality personnel etc.) leading up to the start of fabrication in the various disciplines. The Micron Project team shall review and approve this plan and shall follow up status on a regular basis.
 - 11.9.2 Responsibility for follow up of the above plan lies with the Micron PMO and Project Manager (or designated individual).
- 11.10 Ready for construction
 - 11.10.1 The Project is ready for construction, on a discipline by discipline basis, when all the following elements are in place: (Typical examples)
 - The Engineering is at a mature state, sufficient to be able to issue verified IFC drawings and secure continuous construction.
 - Shop drawings developed from the AFC design drawings, have been issued IFC
 - Sufficient amounts of materials are available on Site

- Workshop space has been allocated and prepared to start construction
- All EHS issues have been sorted out
- A suitable workforce is available that is qualified and has had full EHS Induction Training
- All relevant quality procedures have been issued and approved

11.10.2 Micron PMT will co-operate with xx Contractors name xx in preparations for commencing construction, typically weekly inspections/assessment of any offsite facilities / Procedures /engineering will be performed and a common punch-list listing outstanding issues shall be established. All 'A' punch items shall be closed prior to start of construction.

11.11 Preparedness Audit

11.11.1 In order to confirm the above, formal audits may be carried out of each location and element of the work before any actual construction work can commence. This will normally be a joint audit by Micron and xx Contractor name xx to check that all of the above criteria have been met. A final check-list for use during such audits will be developed and used on a case-by-case basis. The checklist will document the verification activities done by Micron related to construction readiness.

11.12 Design reviews and Risk assessment

11.12.1 The Micron GFTT team, including construction personnel, will attend design reviews and Risk assessment during the detail design phase.

11.12.2 The following engineering design activities are identified as most important focusing on constructability and EHS prior to construction execution:

- Area/Lay-out reviews
- System design reviews
- EPC Planning reviews/analyses covering all project objects.
- Integrated planning reviews, e.g. integrated fabrication and commissioning logic network plan(s) for onshore phase and construction HU & Comm. for offshore phase.
- Constructability / Construction methodology reviews.
- Human Factor and Working environment reviews / analyses done by EPC contractor or others.
- EHS training and mobilization workshops at the site(s).
- Review of design for Commissioning & Start-up

provisions. Commissioning / Start-up risk assessment
Risk assessment.

11.13 Procurement, Material Control and Logistics

11.13.1 Although procurement is represented within the Micron Project team. It is recommended the project team have a dedicated Materials & Logistics Coordinator.

11.13.2 Past experiences has shown that a major contributor to the overall success of a Construction project to ensure that the right materials/equipment arrives at the 'work face' just in time when they are required by the construction workforce.

- If the material/equipment gets there too early, there are risks that they are liable to be damaged or go missing;
- If it arrives too late, there will be schedule delays. There will be inevitable changes to the planned sequence of construction work and it is imperative that the Project team is aware of the material/equipment status at all times.

11.13.3 The Construction Materials & Logistics Coordinator will closely follow up the construction requirements for the next monthly period, considering any new changes to the construction sequence.

- They will determine where material/equipment is located and when it is expected to arrive on site.
- If they can see any shortfalls for the following month, he/she will investigate, along with xx Contractors name xx, any measures that may be available to expedite these shortfalls.
- If this is not possible, they must identify these shortfalls to the Micron /xx contractors xx construction teams who will then have to develop 'work around' solutions and plans.

11.14 Risk Management

11.14.1 Each member of the Micron PMT is responsible for identifying and reporting potential threats and opportunities in the project.

11.14.2 The PMO is nominated the focal point within the Micron PMT for registering input to the risk management process and for ensuring these issues are entered into the Project Risk Register.

11.14.3 Assessment of identified risks and associated mitigating measures will take place monthly, and these meetings will be called by the Project Manager / PMO or his delegate.

11.15 Construction Quality Management

- 11.15.1 Xx Contractors name xx is responsible for ensure all quality requirements are met and shall produce a Quality Plan, Quality procedures, ITP as required to document this.
- 11.15.2 A good relationship with the xx Contractors name xx Quality dept is vital to achieve the quality standard Micron is expecting.
- 11.15.3 A dedicated team of Micron Quality inspectors (If deployed to site) shall closely monitor all aspects of the xx Project name xx construction work. They will work in close cooperation with the Contractors QC group.
- 11.15.4 It is anticipated that the higher Quality standards demanded by this Project may be a challenge to the xx contractors name xx organization.
- 11.15.5 It may be necessary for the Micron discipline leads and Quality team to assist xx contractors name xx rather than just criticize them for any failings.
- 11.15.6 Weekly Inspection Reports / Nonconformity Reporting
 - Quality Lead shall produce a Weekly Surveillance Report (WSR).
 - This report will provide details of activities observed during the week and will highlight any findings relating to quality issues identified. It should also highlight any EHS concerns.
 - If any project team member notices an actual poor-quality situation or sees the potential for a poor-quality situation, he shall raise the observation (suggest creating a Quality Action Issue report (QAI)) for each separate event and registered in a log to be maintained by the QM Lead.
 - QAI should be reviewed immediately with contractor in order to rectify poor quality issues or take steps to prevent poor quality from occurring.
 - A contractor that fails to take immediate action, and/or the situation continues to be an issue, a proper Non-Conformance Report (NCR) shall be raised and the issue be escalated to a higher level within both organizations.

11.16 Clean room Construction Protocol

- 11.16.1 It has been identified by all parties that one of the greatest challenges of this type of Project will be to complete the work to the necessary high standard of cleanliness and dryness demanded by the Project Specifications.
- 11.16.2 The project has developed a Clean room Protocol Procedure (Doc No. xx-xx-xxx) which provides a systematic approach to achieving this standard whilst understanding the different phases that are gone through during the construction process.
- 11.16.3 This Procedure must be fully implemented, and a plan for implementation shall be established. One important aspect will be the full commitment from the entire workforce to achieve this.
- 11.16.4 The Procedure requires a signed declaration by all Sub-contractors and other vendors / companies that their workers will comply. This should be a requirement of all PO packages for sub-contractors and manpower supply companies.

11.17 Construction Completion, Preservation and Handover to Commissioning

11.17.1 Objective

- It is a requirement of the project that full documentation is available to substantiate that all work has been completed to a suitably accepted quality standard and that all equipment functions as intended.
- The process used for documenting this is Construction Completion. This is carried out by the xx contractors name xx Construction group to document that they have completed all their scope of work prior to handing it over to the Commissioning group who will then liven up and test all systems (and areas) prior to their handover to Micron Facilities.
- Within this documentation, there is a requirement to substantiate that all pieces of equipment have been preserved/maintained in accordance with the manufacturer's recommendations.

11.17.2 Completion at the Suppliers

- An integral part of the Construction completion system is the requirement that all Vendor supplied packages follow the same Construction completion system. This means that every valve, instrument, switch, etc. in equipment packages follows the same tag numbering system and routine preservation as the rest of the project.
- At every stage of the package equipment completion, check sheets are filled out to confirm that the package is mechanically complete before the FAT and before shipment to the project site.
- Any incomplete work (punch items) from suppliers shall be completed by the supplier or an organization nominated by the supplier (xx contractor name xx or another local company) in xx country of project xx. Micron project Team will verify that any outstanding work is completed as part of the completion process.

11.17.3 Micron Involvement in completion

- The Micron Project team is responsible for verifying that all activities and work scope items are completed in accordance with the Contract requirements. When all the necessary checks have been carried out, members of the Micron project team and /or Quality team will sign off all necessary check sheets which will document that the work has been satisfactorily completed.
- It is possible that xx contractors name xx will need a lot of help within completion to ensure that nothing is missed or hidden.
- A process will be established to coordinate completion of the construction completion activities, including involvement of Micron personnel.
- The responsibility for ensuring this is in place is given to the PMO.

11.17.4 Handover

- There are two separate Handovers in the project; the first from the Contractor (xx contractors name xx) to the Commissioning group, and the second from Commissioning to Micron Facilities.
- It is important for the Project that this is a seamless transfer and hence there will be Micron project team members proactively working along with the commissioning group, and Facilities personnel working with the Project team and commissioning team at Commissioning Completion.

12.0 Construction Close-Outs

12.1 Completion of Documentation

- 12.1.1 The Construction team will need to document that all construction has been completed and handover the necessary documentation.

12.2 Lessons Learned

- 12.2.1 As with the commencement of this project, Lessons Learned is a very important aspect of Experience Transfer within both the Micron and xx contractors name xx groups.
- 12.2.2 All Micron Project Team personnel will be asked to complete a form which highlights those areas within their own expertise that went well, didn't go well, what should be repeated on new projects and what should not.
- 12.2.3 Feedback on items for improvement on new Projects will be encouraged. These findings will be discussed internally within the Micron project team and with xx contractors name xx before dissemination further within the Micron global system.
- 12.2.4 The PMO / Project Manager is responsible for organizing and ensuring that lessons learned are captured and documented.

13.0 Constructability/Execution Activities

13.1 xx Project name xx Project Phases

- 13.1.1 The following project phases are identified related to constructability / construction preparations and execution: (Typical examples – to be tailored to specific project)
- Detailed design / procurement 30% review
 - Detailed design / procurement 60% review
 - Detailed design / procurement 90% review
 - Detailed design / procurement 100% review
 - Base Build construction plan / sequence
 - MEP plan / sequence
 - Commissioning plan
 - Tool move in
 - Tool Hook up
 - Close out and lessons learned

13.2 Constructability / Construction Execution phase

Table 1 lists and gives status on all key constructability / construction activities for the xx Project name xx contract.

Table 1: Key constructability / construction activities for the xx Project name xx contract

Phase	Contract / Object	Description of pathway action	Responsible	Status / comments
Execution	Fab/ CUB	Prepare and execute a review of contractors "construction method & sub-contracting statement document "including baseline of detail level 5 network plan	Project PM / PMO	
Execution	Fab/ CUB	Identify valuable experiences and lessons learnt from previous projects and transfer / implement those to chosen contractor.	Project PM / PMO	
Design / Execution	Fab/ CUB	Experience transfer is taken care of by establishing frame agreements for critical equipment suppliers	PM / PMO / Procurement	
Design / Execution	Fab/ CUB	Develop EHS training plan / matrix	EHS / PM / PMO	
Execution	Fab/ CUB	Review and comment contractors "constructability plan" that typically shall be presented to Micron during the first 90 days after contract start up.	PM / PMO	
Execution	Fab/ CUB	Develop contract objects discipline checklist / key words for constructability review. This can be done as a joint effort with contractor.	PM / PMO	
Execution	Fab/ CUB	Establish attendance plan / matrix for (Typical examples) - Area layout and system design reviews - HAZOPS during detail design - Human factor/ working environment reviews - Constructability/construction methodology reviews - Engineering/ procurement/ Construction planning reviews - EHS training and mobilization and mobilization workshops at site	PM / PMO	

Phase	Contract / Object	Description of pathway action	Responsible	Status / comments
		- Review of Commissioning and start up provisions - Commissioning/ start up risk assessment HAZOP		
Execution	Fab/ CUB	Carry out constructability reviews together with Contractor	PM / PMO	
Execution	Fab/ CUB	Update / establish a site-specific Construction Execution plan for the Project	PM / PMO	
Execution	Fab/ CUB	Establish and implement a construction readiness check sheet	PM / PMO	
Execution	Fab/ CUB	Together with contractor establish a procedure and schedule for implementation of Cleanroom protocol	PM / PMO	
Execution	Fab/ CUB	Review and accept fabrication method sequence in conjunction with baseline update	PM / PMO	

14.0 Document Control

14.1 Approval: FCG_ENG_CONSTRUCTION_APPR

14.2 Notification:

- GLOBAL_EHS
- GLOBAL_EHS_SAFETYHEALTH
- GLOBAL_FAC_COMMUNICATIONS
- GLOBAL_FAC_CONST_SOP
- GLOBAL_FAC_MANAGERS
- GLOBAL_FAC_NOTIFY
- GLOBAL_SECURITY_TEAM
- GP_INDIRECT_FACILITIES
- FCG_F4_FAC_NOTIFY
- FCG_F6_FAC_NOTIFY
- FCG_F10_FAC_NOTIFY
- FCG_F11_FAC_NOTIFY
- FCG_F15_FAC_NOTIFY
- FCG_F16_FAC_NOTIFY
- FCG_MMP_FAC_NOTIFY
- FCG_MMY_FAC_NOTIFY
- FCG_MSB_FAC_NOTIFY
- FCG_MTB_FAC_NOTIFY
- FCG_MXA_FAC_NOTIFY
- WW_FACILITIES_MANAGERS_APPR

14.3 Final Viewer Access: All Facilities Departments

14.4 Retention: Adherence to the requirements specified in the [Global Facilities - Document Control Procedures](#) and [Global Quality Records Retention Standard](#)

14.5 Review: Biennial

15.0 Revision History

Revision #	Section Changed	Details of Changes	Date
0	N/A	New Document	11/26/2019
1	Title	Changed document title from Global Facilities – Construction Execution Plan -XXX - SITE to Global Facilities - Design & Construction Standard - Construction Execution Plan	11/24/2021
	Section 4	Added IIF	
	Section 6	Sub-Section 6.2.1: Added The "xx" and supporting	
	Section 7	Sub-Section 7.2: Updated Figure 1 Sub-Section 7.3.6: Added NDE (Non-destructive examination) of welds	
	Section 14	Updated Approval Group to FCG_ENG_CONSTRUCTION_APPR Updated the Notification Groups to GLOBAL_EHS, GLOBAL_EHS_SAFETYHEALTH, GLOBAL_FAC_COMMUNICATIONS, GLOBAL_FAC_CONST_SOP, GLOBAL_FAC_MANAGERS, GLOBAL_FAC_NOTIFY, GLOBAL_SECURITY_TEAM, GP_INDIRECT_FACILITIES, FCG_F4_FAC_NOTIFY, FCG_F6_FAC_NOTIFY, FCG_F10_FAC_NOTIFY, FCG_F11_FAC_NOTIFY, FCG_F15_FAC_NOTIFY, FCG_F16_FAC_NOTIFY, FCG_MMP_FAC_NOTIFY, FCG_MMY_FAC_NOTIFY, FCG_MSB_FAC_NOTIFY, FCG_MTB_FAC_NOTIFY, FCG_MXA_FAC_NOTIFY, WW_FACILITIES_MANAGERS_APPR	
2	Section 2	2.1 Updated	03/14/2023
	Section 3	Updated: GFTT Added: FCT, GFC, Global Facilities Central Team, Global Facilities Construction Team, Global Facilities Operations Support Team,	

		Global Facilities Planning Team	
	Section 5	5.3.3 Item 5 Updated	
	Section 7	7.2.4 Updated 7.2.7 Updated 7.3 Updated Figure 1: Standard Project Organization Structure Updated	
	Section 8	8.1.1 Updated 8.1.3 Updated	
	Section 11	11.10.1 Item 1 Updated	